

Abdelwahab Mahmoud

Senior Software Engineer · Backend & Platform

Cairo, Egypt

+201147784094 | abdelwahabahmed93@gmail.com | www.abdelwahab.dev | github.com/abdelwahab313 | linkedin.com/in/abdelwahab313

Summary

Senior backend engineer with 8+ years building and owning production Ruby on Rails services used by real customers, from the multi-provider LLM engine to the AWS platform it runs on. I work backend-first across the full stack: highly scalable, performant Rails and Node.js services, asynchronous queueing and streaming, React when needed, Datadog and OpenTelemetry for monitoring, and data-intensive paths on Redis, MySQL, and PostgreSQL. I ship features end-to-end; when production breaks, I find the root cause and fix it, then harden the path so the failure does not repeat.

Work Experience

OnTheGoSystems

Remote (Hong Kong)

Senior Software Engineer

Dec. 2024 - Present

- Built the multi-provider Rails LLM engine (Bedrock, OpenAI, Gemini) with provider fallback, JSON repair, Langfuse tracing and rate-limit retries covering the errors behind 65% of review failures.
- Shipped an agent-driven triage platform solo in 12 weeks; its pass/fail call lives in code, not the prompt.
- Built the production tooling AI agents run behind: read-only access, rollback wrapper, audit log and a human-only write gate.
- Migrated the product onto ECS Fargate, then owned the platform and its Terraform for 16 months, running 90% on Spot.
- Replaced ECS autoscaling with a queue-depth Lambda autoscaler that drains long-running jobs instead of killing them.
- Traced Aurora MySQL at 100% CPU to a single query plan scanning 7.33M rows per call, then throttled it from 1,013 calls to about 100.
- Owned credits and billing correctness for a metered LLM product; designed a unified authorization architecture to replace scattered checks (4 ADRs, first step merged).
- Wrote the ISO 27001 audit checklist and built the DR and data-loss-prevention programs to a 30-minute RTO and 15-minute RPO.

Dailymealz

Riyadh, Saudi Arabia

Senior Software Engineer

Oct. 2021 - Dec. 2024

- Built the real-time order service behind a 100,000 daily-user delivery product, on Node.js and Socket.IO, pushing kitchen, driver and delivery status to mobile and web clients.
- Decomposed it out of the Laravel monolith into a standalone service in an event-driven, Service-Oriented Architecture (SOA), exchanging state changes over SQS with a REST API contract.
- Served hot order state from Redis with MySQL behind it, keeping status fan-out off the transactional path.
- Migrated the service from JavaScript to TypeScript.
- Led a team of 5 building a driver location check-in system: planning, reviews and pairing across the driver app.
- Instrumented order, driver and kitchen flows with Datadog and OpenTelemetry, cutting Mean Time To Resolve 30%.

Nformacy

Cairo, Egypt

Full-Stack Developer

Oct. 2020 - Dec. 2021

- Led features from business requirements to deployment, backend and frontend, and moved the web apps to a SaaS model.
- Built mentor booking with Zoom, ClickUp and calendar integrations; wrote BDD tests and managed CI/CD pipelines.

DevSquads (formerly Fikr Labs)

Cairo, Egypt

Software Engineer

Jan. 2018 - Oct. 2020

- Built software in self-organizing Extreme Programming teams with TDD, thin vertical slices and test-covered refactoring of legacy code, across React Native, Ruby on Rails, Java Spring and React.
- Mentored teams adopting XP and Agile practices: pairing, code review discipline, iterative planning.

Skills

Programming Languages

Ruby, TypeScript, JavaScript, Java (Spring), Python, SQL, PHP

Backend Development

Ruby on Rails, Node.js, Express, Socket.IO, REST API design, service-oriented architecture, asynchronous messaging (SQS), background workers, FastAPI

Frontend Development

React, Next.js, Redux, Hotwire (Turbo & Stimulus), React Native, Tailwind CSS

AI & LLM

AWS Bedrock, OpenAI, Gemini, agentic workflows, Claude Agent SDK, MCP, Langfuse, evaluation harnesses

Cloud & Infrastructure

AWS (ECS Fargate, Lambda, SQS, Step Functions, CloudWatch, S3, IAM), Terraform, Docker, GitLab CI

Observability

Datadog, OpenTelemetry, AWS CloudWatch, Langfuse tracing

Databases

PostgreSQL, MySQL & Aurora, Redis, DynamoDB

Testing

TDD (Test-Driven Development), RSpec, Capybara, Playwright, Cypress, Jest, VCR

Spoken Languages

English, Arabic (native), German (beginner)

Education

Faculty of Science Cairo University

Bachelor of Science in Geophysics

Cairo, Egypt
Sep. 2012 - Jun. 2016